AI-POWERD CONGNITIVE SCREENING AND ANALYSIS PLATFORM

25-26J-439

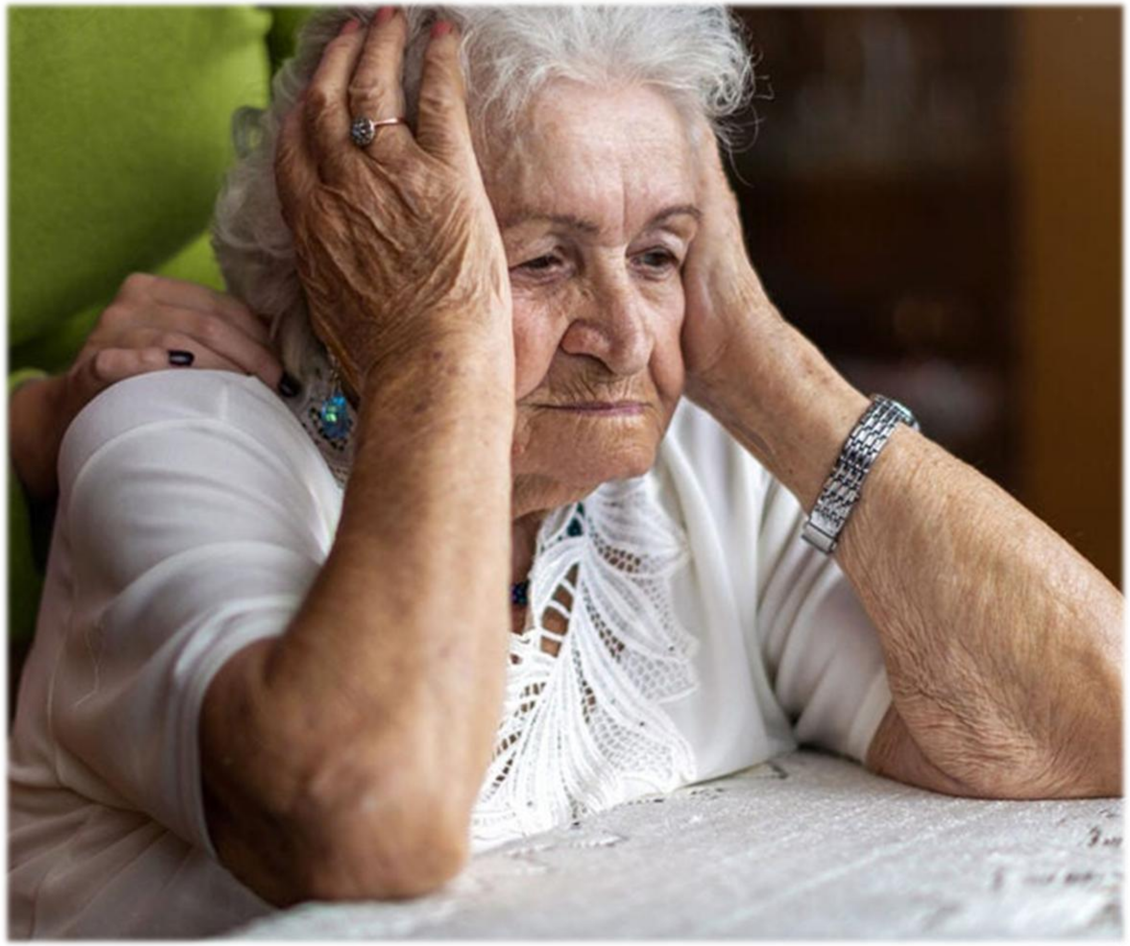
CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Problem Background

- Dementia: 55M+ cases worldwide, 10M new annually (WHO 2021).
- Sri Lanka: rising elderly population, limited affordable screening tools.
- **Early detection is critical** to slow progression and improve quality of life.
- Regular hospital visits are **inconvenient** due to transport issues and time burden.

Impact:

Early detection enables timely treatment and support, improving the patient's quality of life while easing the emotional and practical burden on caregivers.



CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

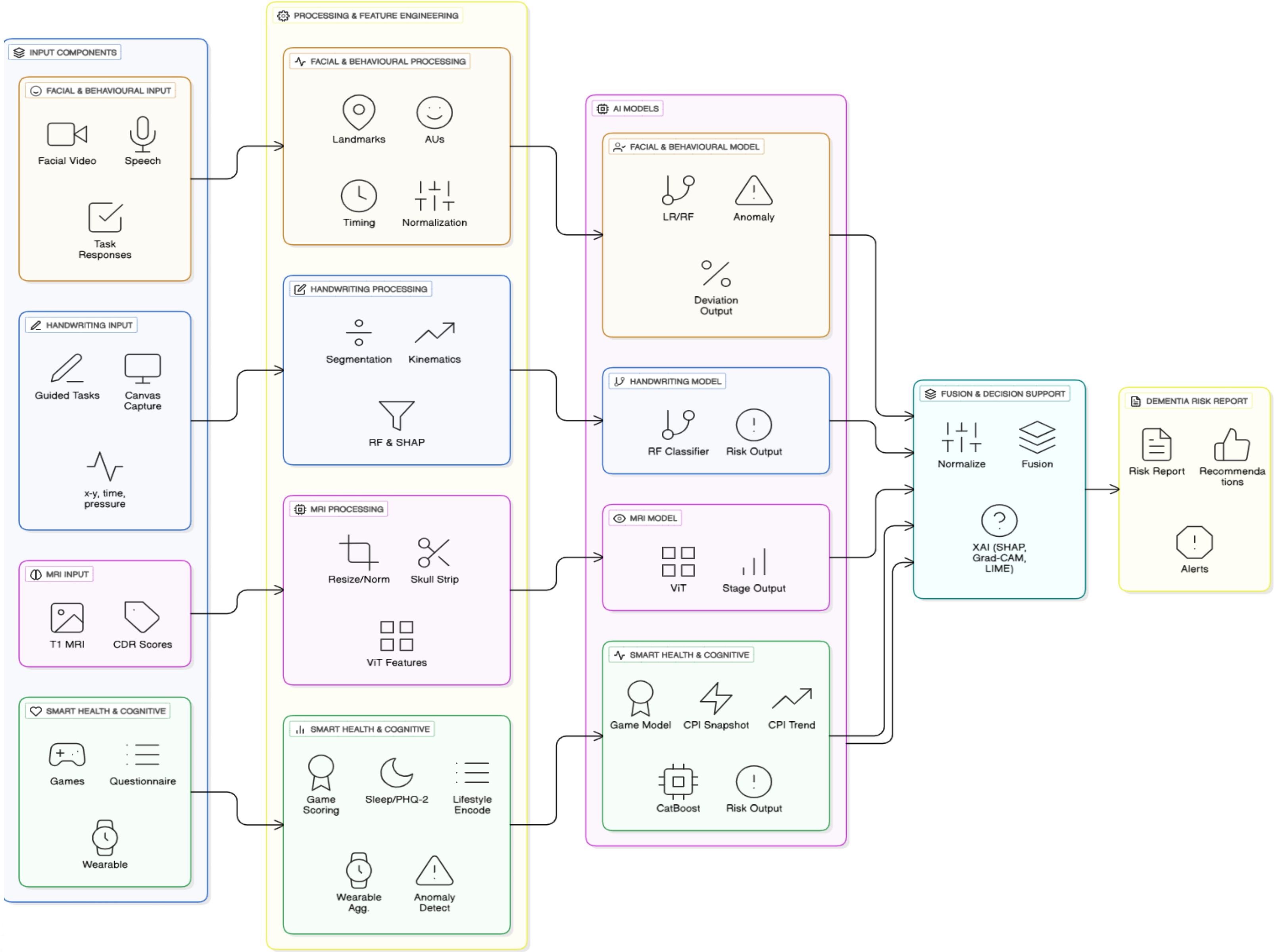
Target Users

- Elderly population (primary)
- Healthcare professionals (support tool)
- Caregivers & clinics
- Low-resource & remote communities

What is new?

- Multimodal fusion of speech & behavioural cues
- Non-invasive and cost-effective screening
- Screening-oriented (not diagnostic)
- Designed for real-world accessibility

Design Excellence of the Overall Solution



IT22358998 | SENARATNE H.D.

Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Software Engineering

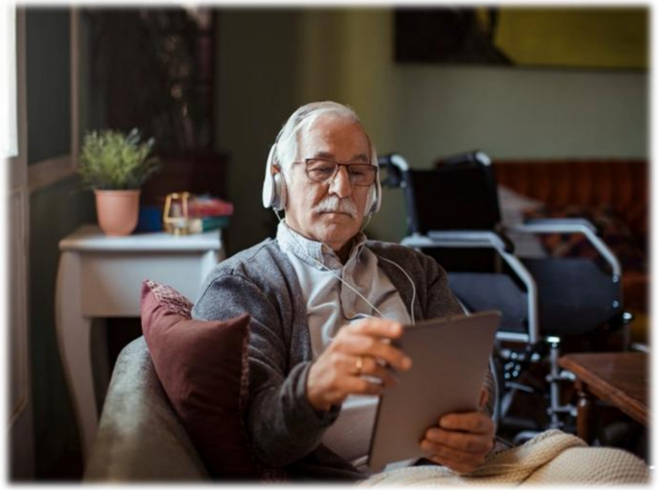
Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Problem

- Relies on expensive hardware, limiting accessibility & scalability
- Requires **in-person clinical visits**, which are time-consuming and inconvenient
- Do **not explain why** a prediction was made

Solution

- Captures handwriting **directly via a web browser** using HTML Canvas
- No special hardware required (works on laptop/touchscreen devices)
- Guides users through **handwriting tasks**
- Automatically extracts movement-based features
- Generates a **cognitive risk score** with **explainable results**



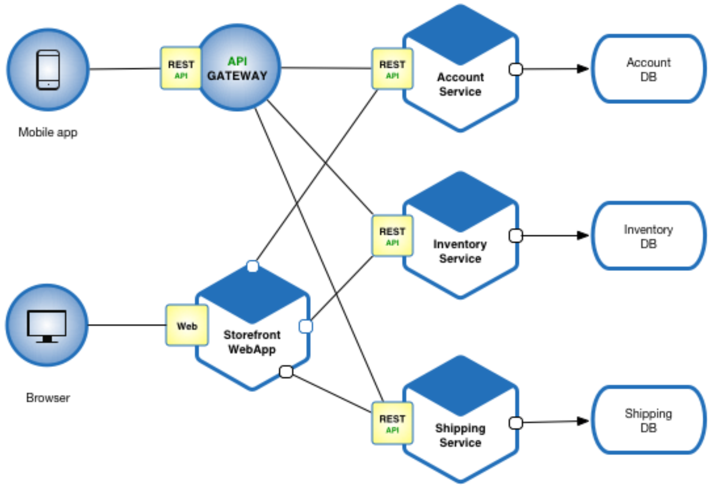
Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

User Requirements Addressed

- **Accessibility** → Works on common devices without clinical equipment
- **Ease of Use** → Step-by-step guided handwriting tasks
- **Time Efficiency** → Short assessment suitable for home use
- **Clarity of Results** → Risk score with simple explanations

Design Excellence & Research Contribution

- Modular **microservice architecture** separating frontend and ML services
- Explainable results using SHAP
- Accessible, browser-based screening
- Scalable foundation for future multimodal cognitive assessment



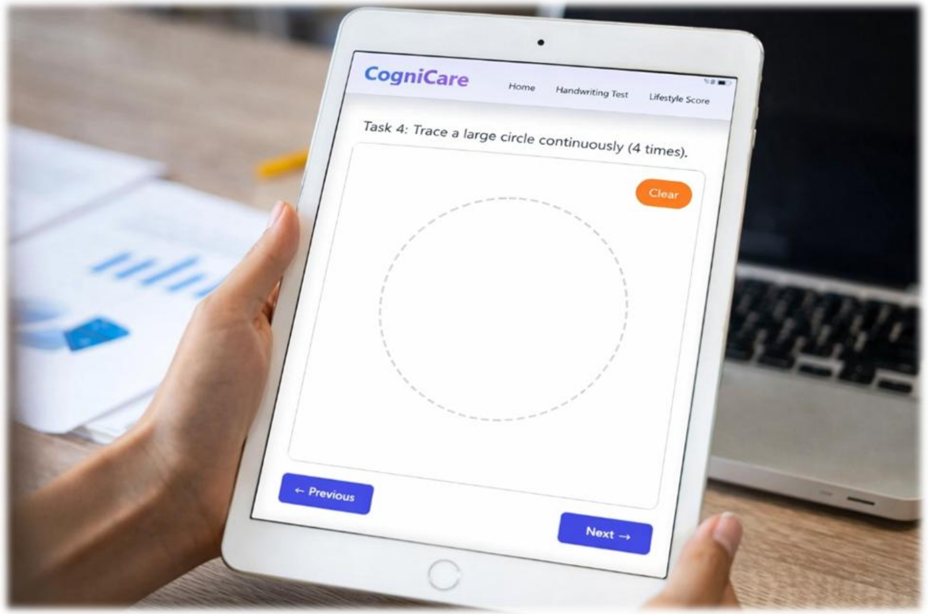
Handwriting-Based Dementia Screening Tool for Mobile and Web Platforms

Observed Feedback

- Users found the handwriting **tasks easy to understand**
- Browser-based input **felt less intimidating** than clinical tests
- Explainable results **improved user confidence and trust**
- Feedback highlighted **clear instructions, visual guidance, and progress tracking.**

Actions Taken

- Improved UI styling and clarity
- Added task progress indicators
- Simplified feature explanations for non-technical users



IT22901644 | DISSANAYAKE M.P.S.

Dementia stage classification from MRI using Vision Transformer and explainable AI.

Software Engineering

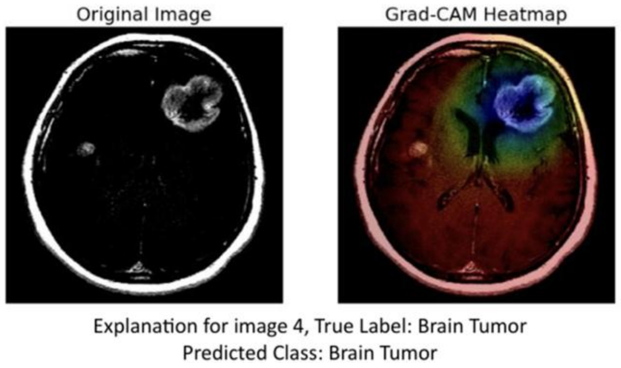
Dementia stage classification from MRI using Vision Transformer and explainable AI.

Sub - Problem:

- Early and accurate identification of dementia stages using MRI scans is challenging due to subtle structural brain changes and the lack of explainable AI support for clinicians. Manual interpretation is time-consuming and subjective.

Solution:

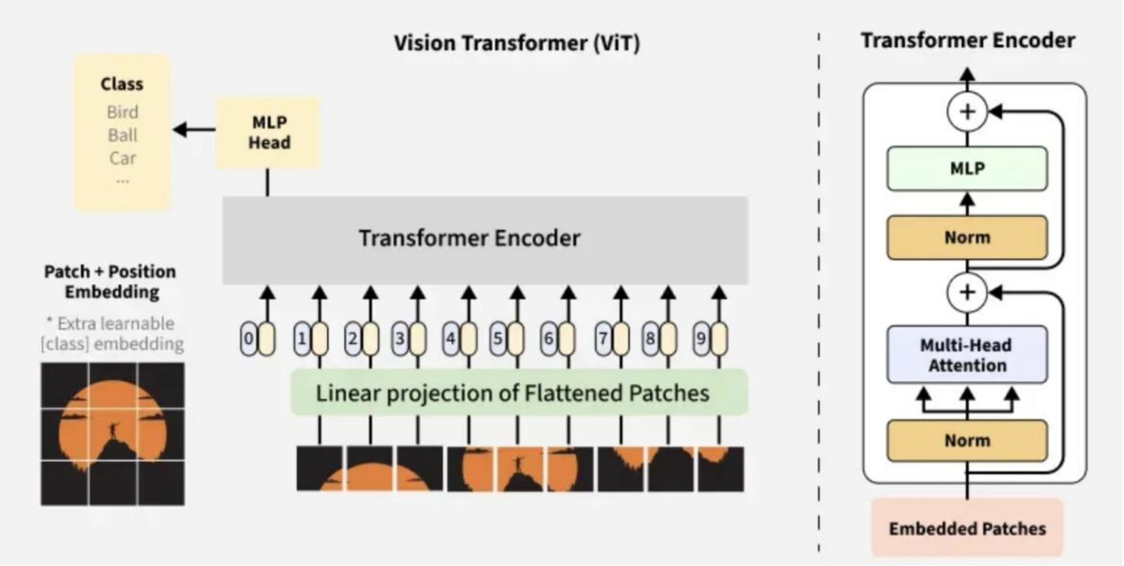
- Automatically classifies MRI brain images into four dementia stages aligned with the Clinical Dementia Rating (CDR) scale
- Uses a Vision Transformer (ViT) model to capture global structural patterns in brain MRIs
- Provides prediction confidence scores to indicate model reliability
- Generates Grad-CAM heatmaps to visually highlight brain regions influencing predictions



Dementia stage classification from MRI using Vision Transformer and explainable AI.

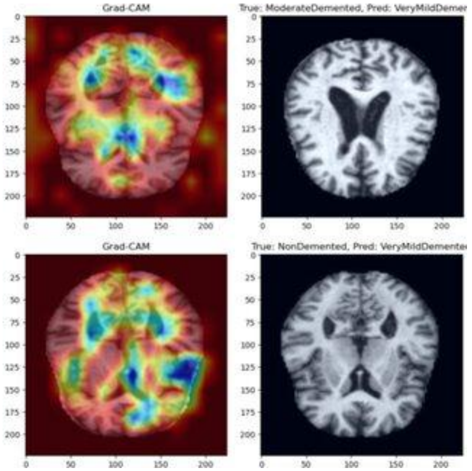
User Requirements Addressed:

- Clinicians → Accurate stage classification
  Explainable visual evidence (heatmaps)
- Caregivers / Patients → Clear stage interpretation
  Easy-to-use interface
- System Requirements → Fast inference via API
  Secure MRI image handling



Design Excellence / Contribution

- **Explainable AI Integration:**
  Grad-CAM adds transparency by visually mapping influential brain regions
- **Modern Deep Learning Architecture:**
  Vision Transformer outperforms traditional CNNs for global MRI pattern learning
- **Microservices Architecture:**
  Separate ML service (FastAPI) + backend gateway (Node.js) ensures scalability
- **Clinical Alignment:**
  Output classes mapped directly to the **CDR dementia staging framework**



Dementia stage classification from MRI using Vision Transformer and explainable AI.

User Feedback on Prototype

•**Medical Perspective (Expected):**
- Heatmaps improve trust and interpretability
- Confidence scores support decision-making

•**Usability Feedback:**
- Simple MRI upload process
- Clear visualization of predictions

•**Improvements Identified:**
- Further refinement of Grad-CAM resolution
- Calibration of confidence scores for clinical robustness

IT22358820 | PIRUTHIVI K

SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Software Engineering

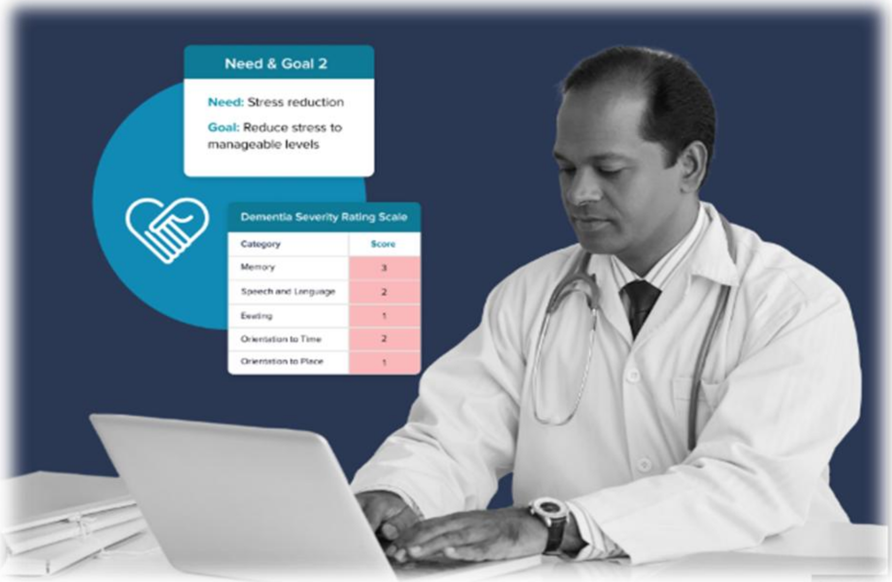
SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

Sub-Problem Addressed

- Rely on **clinical, paper-based tests**
- Do not support **continuous monitoring**
- Do not integrate **cognitive, lifestyle, and wearable data**
- Are **costly**, clinic-dependent, and not scalable for early screening

This solution

- low-cost.
- AI-powered digital platform that integrates cognitive game performance, lifestyle profiling, and smartwatch-based health monitoring into a unified system.
- explainable screening and Recommendations **.**



SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

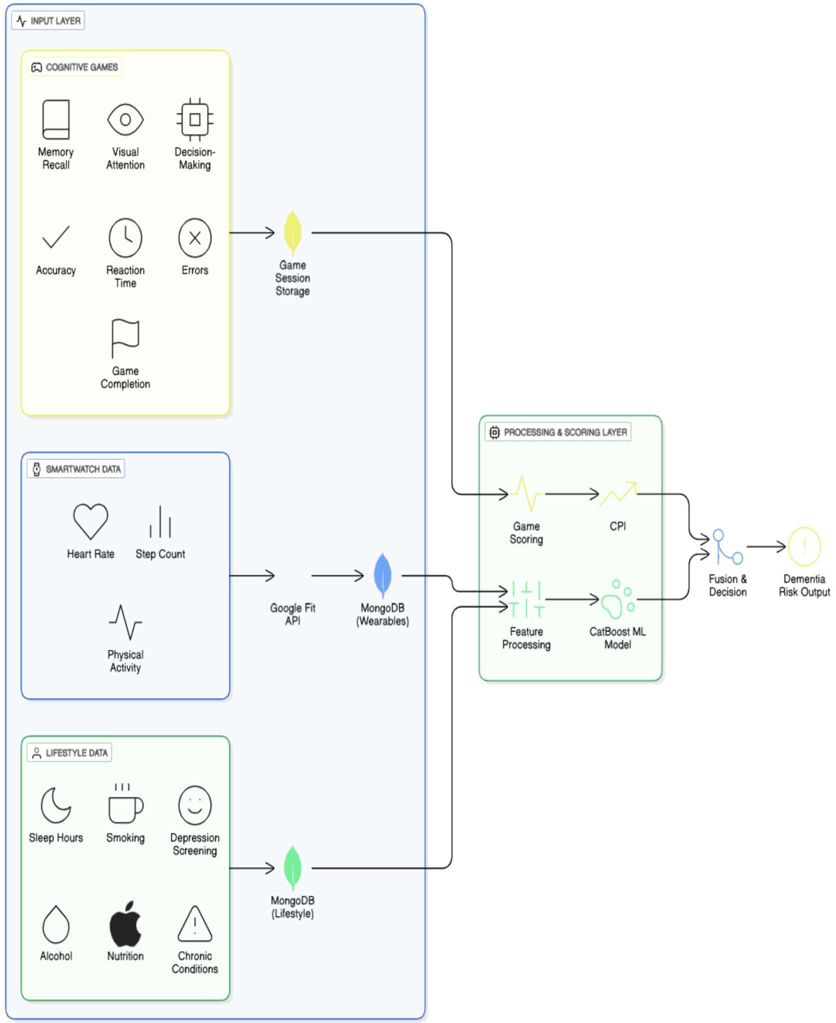
User Requirements Addressed

Low-cost and accessible: Uses smartphones, smartwatches, and a web interface without clinical hardware.

Elderly-friendly and remote: Simple cognitive games and questions enable home-based screening.

Immediate and continuous feedback: Provides instant results and long-term cognitive monitoring.

Clear and actionable results: Doctor-friendly summaries support reassessment and early intervention.



Design Excellence & Technical Contribution

Microservice design: React frontend, Node/Express backend, Python (FastAPI) ML service

Feature Engineering & ML

lifestyle, and wearable features (Google Fit) engineered and modeled

Clinical dementia risk prediction using CatBoost

Wearable anomaly detection prepared as future enhancement

Cognitive Scoring & Risk Logic

Per-game scoring with Snapshot CPI and Trend CPI separation

Interpretable, rule-based fusion for ethical screening

SMART HEALTH MONITORING AND DEMENTIA RISK ANALYSIS

User Feedback on Prototype

- Game-based tests were **more engaging** than questionnaires
- CPI-based summaries were **easy to understand**
- Technical reviewers appreciated:
    - Low-cost and scalable design
    - Transparent fusion logic

IT22888952 | NATHEER H A H

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASK BEHAVIOUR

Software engineering

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASKBEHAVIOUR

Sub-Problem Addressed

Traditional dementia screening approaches:

- Focus mainly on **clinical tests and MRI scans**
- Are **clinic-dependent** and expensive
- Rely on **single modalities** (speech only / MRI only)
- Do not capture **natural behavior during daily tasks**
- Provide **limited explainability** for screening decisions

Solution:

- Low-cost, **non-invasive**, AI-based screening approach
- Integrates **speech patterns**, **facial expressions**, and **task behavior**
- Enables **home-based, early dementia risk screening**
- Provides **explainable cognitive deviation indicators**

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASKBEHAVIOUR

User Requirements Addressed

Low-cost and accessible:

- Uses microphone, camera, and web/mobile interface
- No clinical equipment or wearable dependency required

Elderly-friendly and remote:

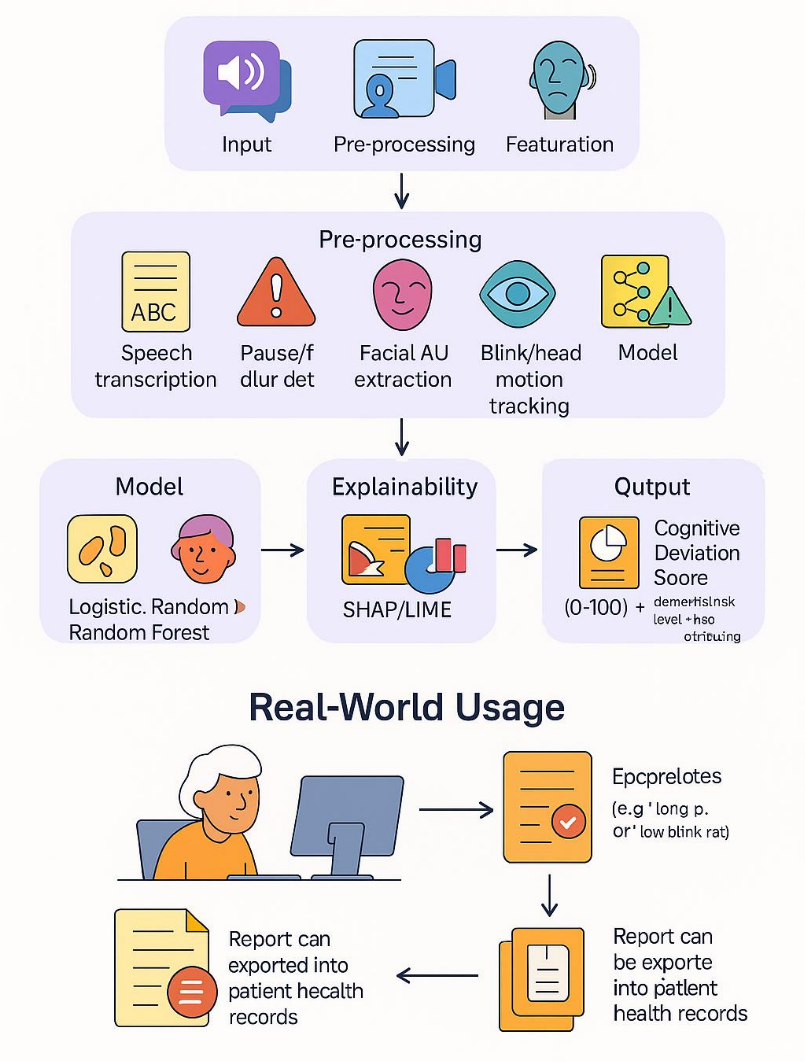
- Natural speech and simple task interactions
- No medical terminology or complex instructions

Early and continuous screening:

- Can be repeated periodically
- Supports monitoring subtle cognitive changes over time

Clear and actionable results:

- Produces an **explainable Cognitive Deviation Score**
- Helps clinicians and caregivers understand *why* risk was flagged



Design Excellence & Technical Contribution

Multimodal AI Architecture

- Speech feature analysis (pauses, fillers, lexical richness)
- Facial expression & behavioral cue extraction
- Task-based interaction signals combined into a unified score

Explainable Screening Logic

- Uses interpretable ML models
- SHAP / LIME used to explain feature influence

Scalable Microservice Design

- React frontend for interaction
- Python (FastAPI) ML inference service
- Backend integration ready for full CogniCare platform

DEMENTIA SCREENING VIA SPEECH, FACIAL EXPRESSIONS, AND TASKBEHAVIOUR

User Feedback on Prototype

Feedback from peer users and evaluators showed:

- Speaking naturally felt **less stressful** than answering formal questionnaires
- Users were comfortable with **short speech tasks** rather than long interviews
- Facial expression–based analysis was perceived as **non-intrusive**
- Simple task-based interactions were easier than clinical-style tests

Technical reviewers appreciated:

- **Low-cost, home-friendly design** without clinical equipment
- **Multimodal integration** of speech, facial, and behavioral cues
- **Explainable outputs** that clearly show contributing risk factors

CogniCare: An AI-Powered Cognitive Screening & Analysis Platform

Customer Persona

Patient

Elderly Individual (60+)

- Needs early, non-invasive cognitive screening & analysis
- Limited technical skills
- Concerned about privacy & stigma
- Benefits from simple, accessible assessments

Caregiver

Family Member / Caregiver

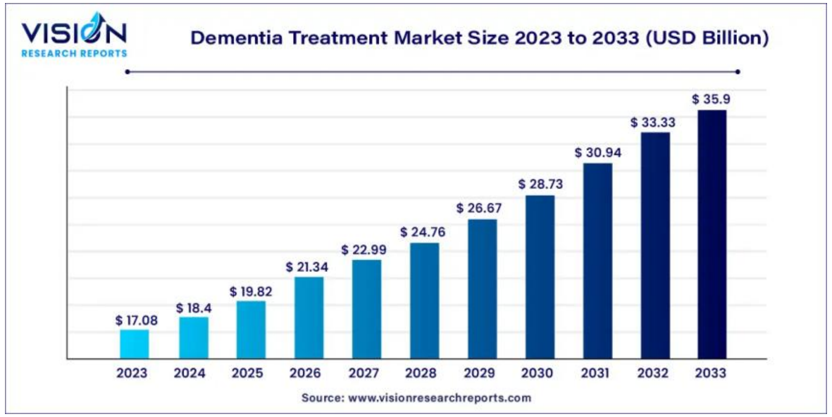
- Monitors cognitive changes over time
- Needs clear risk insights and easy-to-understand explanations
- Wants affordable, home-friendly tools
- Supports medical decision-making

Clinician

Doctor / Neurologist

- Requires objective screening support
- Limited consultation time
- Needs explainable AI outputs (SHAP, reports)
- Uses system as decision-support, not diagnosis

CogniCare: An AI-Powered Cognitive Screening & Analysis Platform



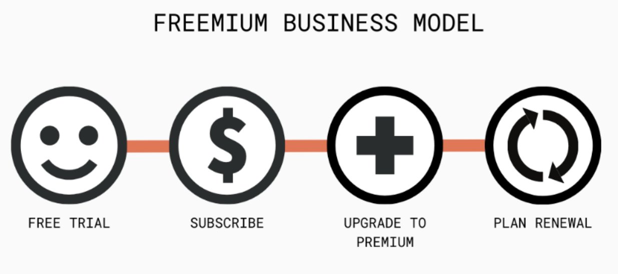
Market Size

- 55+ million people worldwide live with dementia (WHO)
- 10 million new cases annually due to ageing population
- Strong need for **low-cost, remote early screening**, especially in LMICs

What is Unique in Your Solution

- **Unified Multimodal Platform** combining handwriting, MRI, facial/behavioural, and wearable data.
- **Two-Stage AI Pipeline**: Demented vs Non-Demented → dementia stage prediction.
- **Explainable AI (Grad-CAM)** providing visual evidence for clinical trust.
- **Localized & Accessible Design** with Sinhala support and mobile/web deployment.

How Will the Cost Be Recovered?



- Low Initial Cost using open-source tools and free public datasets Project proposal
- Freemium Model: basic screening free, advanced analytics via subscription.
- Institutional Licensing for hospitals, clinics, and research organizations.
- Partnerships & Scalability through NGOs and cloud-based deployment.